

IT 213: Midterm Study Guide (Expanded)

Concordia University of Edmonton

Instructor: Bernal Manzanilla

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Abstract

This guide summarizes the core concepts, protocols, and terminal commands covered in Weeks 1 through 6. Focus on understanding how these systems interact, the historical context of communication, the analogies discussed in class, and the outputs of your lab tools.

1 Week 1-2: Foundations of Communication

1.1 History: The Telegraph & Physical Limits

To understand modern networks, we look at the first digital network: the Transatlantic Telegraph Cable.

- **The Problem of Dispersion:** When sending electrical pulses (dots and dashes) over massive distances, the signal degrades and "spreads out." If you send pulses too closely together, they smear into one continuous blur. This is called **Dispersion**.
- **Inter-Symbol Rate (Baud Rate):** Because of dispersion, there is a physical speed limit to how fast you can change the state of the signal (e.g., from voltage high to voltage low) without causing **Inter-Symbol Interference (ISI)**. The Inter-Symbol rate is the number of symbol changes per second.

1.2 Sharing the Wire: Multiplexing

How do multiple users share a single cable without their signals crashing into each other? We use **Multiplexing**.

Type	How it Works	Real-World Analogy
FDM (Frequency Division)	The cable is divided into different frequency bands. Each user gets their own dedicated frequency 100% of the time.	Radio Stations: 91.7 FM and 100.3 FM exist in the air at the same time, but on different frequencies.
TDM (Time Division)	Users take turns. Each user gets the entire frequency band, but only for a fraction of a second.	A Panel Discussion: Everyone shares one microphone, but each person gets 30 seconds to speak.

1.3 Circuit Switching vs. Packet Switching

- **Circuit Switching (Old Telephone Networks):** Uses FDM/TDM to reserve a dedicated path for the entire conversation. If you are silent, the wire is wasted.
- **Packet Switching (The Modern Internet):** Data is broken into chunks (packets). Everyone shares the wire equally. If the wire is busy, your packet waits in a queue (a buffer).

2 The Protocol Stack & Encapsulation

Data doesn't just fly through the air; it gets wrapped in layers. Think of this as the **Russian Doll** concept (Encapsulation). Every time data goes down a layer, a new header is added.

Memorize: The Data Units (Packages)

Different layers call the data package by a different name. You must know these!

- **Layer 7 (Application):** The package is called a **MESSAGE**. (e.g., HTTP, DNS).
- **Layer 4 (Transport):** The package is called a **SEGMENT**. (e.g., TCP, UDP).
- **Layer 3 (Network):** The package is called a **DATAGRAM**. (e.g., IP, ICMP).
- **Layer 2 (Link):** The package is called a **FRAME**. (e.g., Ethernet, Wi-Fi).
- **Layer 1 (Physical):** The package is just raw **BITS** (0s and 1s).

3 Layer 7: Application (HTTP & DNS)

3.1 HTTP (HyperText Transfer Protocol)

HTTP is the text-based language of the web. It is **stateless** (it forgets you immediately).

- **GET Request:** "Give me a file." Data is visible in the URL. Safe and bookmarkable.
- **POST Request:** "Take this data." Used for passwords/forms. Hidden in the message body.
- **Status Codes:** **200 OK** (Success), **404 Not Found** (Client error), **500 Internal Error** (Server code crashed).

3.2 DNS (Domain Name System)

DNS is the internet's **Phonebook**. It translates human names (**google.com**) into computer numbers (IP Addresses).

- **A Record:** Maps a Name directly to an IPv4 Address.
- **CNAME Record:** Maps a Nickname to a Real Name (Canonical name).
- **MX Record:** Points to the Mail server.

4 Layer 4: Transport (TCP vs. UDP)

To deliver data to the correct application, the Transport layer uses **Ports** (The "Apartment Number"). Memorize these Celebrity Ports: **80 (HTTP)**, **443 (HTTPS)**, **53 (DNS)**, **22 (SSH)**.

Feature	UDP (User Datagram Protocol)	TCP (Transmission Control Protocol)
Header Size	8 Bytes (Tiny)	20 Bytes (Large)
Reliability	Unreliable ("Fire and Forget")	Reliable (Retransmits lost packets)
Connection	Connectionless	3-Way Handshake required
Best For	Streaming, DNS, fast queries	Web browsing, Email, File downloads

4.1 Inside TCP

- **Sequence Numbers:** TCP tracks data by counting **Bytes**, not packets!
- **Window Size:** Controls flow. It tells the sender: "I only have room for X bytes, don't overwhelm my buffer."
- **The 3-Way Handshake:** SYN (Client) → SYN-ACK (Server) → ACK (Client).

TCP Flags (The Control Panel)

- **[S] SYN:** Start a connection.
- **[.] ACK:** Acknowledge receipt of data.
- **[F] FIN:** Finish/Close connection politely.
- **[R] RST:** Reset! The emergency brake (connection refused/firewall blocked).
- **[P] PSH:** Push! "Don't buffer this, send to the application immediately" (The impatient waiter).

5 Layer 3: Network (ICMP & Distances)

We measure distance on the internet in **Time (ms)** and **Hops (Routers)**. We use **ICMP** (Internet Control Message Protocol) to diagnose the network.

- **Ping:** Measures RTT (Round Trip Time). Sends an *Echo Request (Type 8)*, receives an *Echo Reply (Type 0)*.
- **TTL (Time to Live):** A self-destruct timer on a packet to prevent infinite loops.
- **Traceroute:** Maps the path to a server by intentionally sending packets with short TTLs ($TTL = 1$, then 2, etc.) to force routers to send back "Time Exceeded" errors.

6 The Command Line Toolkit

You must know what tool to use for a given problem.

`nslookup gaia.cs.umass.edu`

Queries the DNS phonebook to find the IP address of a server.

`ping gaia.cs.umass.edu`

Checks if a server is alive and measures the Round Trip Time (RTT).

`tracert (Win) / traceroute (Mac)`

Lists every router (hop) between you and the destination.

`curl http://site.com/file.txt -o file.txt`

Acts like a text-based web browser. The `-o` flag saves the output to your hard drive instead of cluttering the screen, and shows download statistics.

`sudo tcpdump -nv port 80 > log.txt`

Sniffs raw network packets (requires admin/sudo). We redirect it (`>`) to a text file for analysis.

`grep "Flags \[S\]" log.txt`

Searches massive text files for specific Regular Expressions (Regex). Here, we escape the brackets with a backslash to find the literal `[S]` (SYN) flag.